

Name:

Reg. No:

APJ ABDULKALAM TECHNOLOGICAL UNIVERSITY
SECOND SEMESTER M.TECH. DEGREE EXAMINATION SEPTEMBER 2023

Course Code: 222ECS016

Course Name: KNOWLEDGE BASED SYSTEM DESIGN

Max. Marks: 60

Duration: 2.5 hours

Branch: Computer Science and Engineering

PART A

(Answer all questions in PART A. Each question carries 5 marks)

Q. No.		Marks
1	Discuss the different types of knowledge.	(5)
2	With an example, illustrate the working of a heuristic classification method.	(5)
3	Explain briefly the difficulties with the KBS.	(5)
4	What is the need of a distinct knowledge structure for knowledge representation? How it differs from commonly used data structures?	(5)
5	Write a short note on semantics of expert systems.	(5)

PART B

(Answer any five full questions in PART B. Each full question carries 7 marks)

6	a. Explain the key components of knowledge.	(3)
	b. Give the general structure of knowledge-based system. Why is knowledge base considered as key component of knowledge-based system?	(4)
7	a. What are the techniques that can be used to eliminate irrelevant parts of the search space to quickly find optimal solutions?	(3)
	b. Explain a search space approach for solving 8-puzzle problem.	(4)
8	a. Differentiate between forward chaining and backward chaining.	(3)
	b. Using appropriate assertions and rules, illustrate the working of forward chaining technique in a medical diagnosis domain.	(4)

- 9 a. Illustrate the working of inference engine in a KBD. (3)
- b. With examples, explain how inference engine infers new knowledge as and (4)
 when needed.
- 10 With a neat diagram, explain the development methodology for a knowledge- (7)
 based system.
- 11 Illustrate different methods for representing procedural knowledge. Give (7)
 examples.
- 12 a. Differentiate between case-based reasoning and rule-based reasoning. (3)
- b. With an example, justify which one will be suitable for designing knowledge (4)
 based systems.